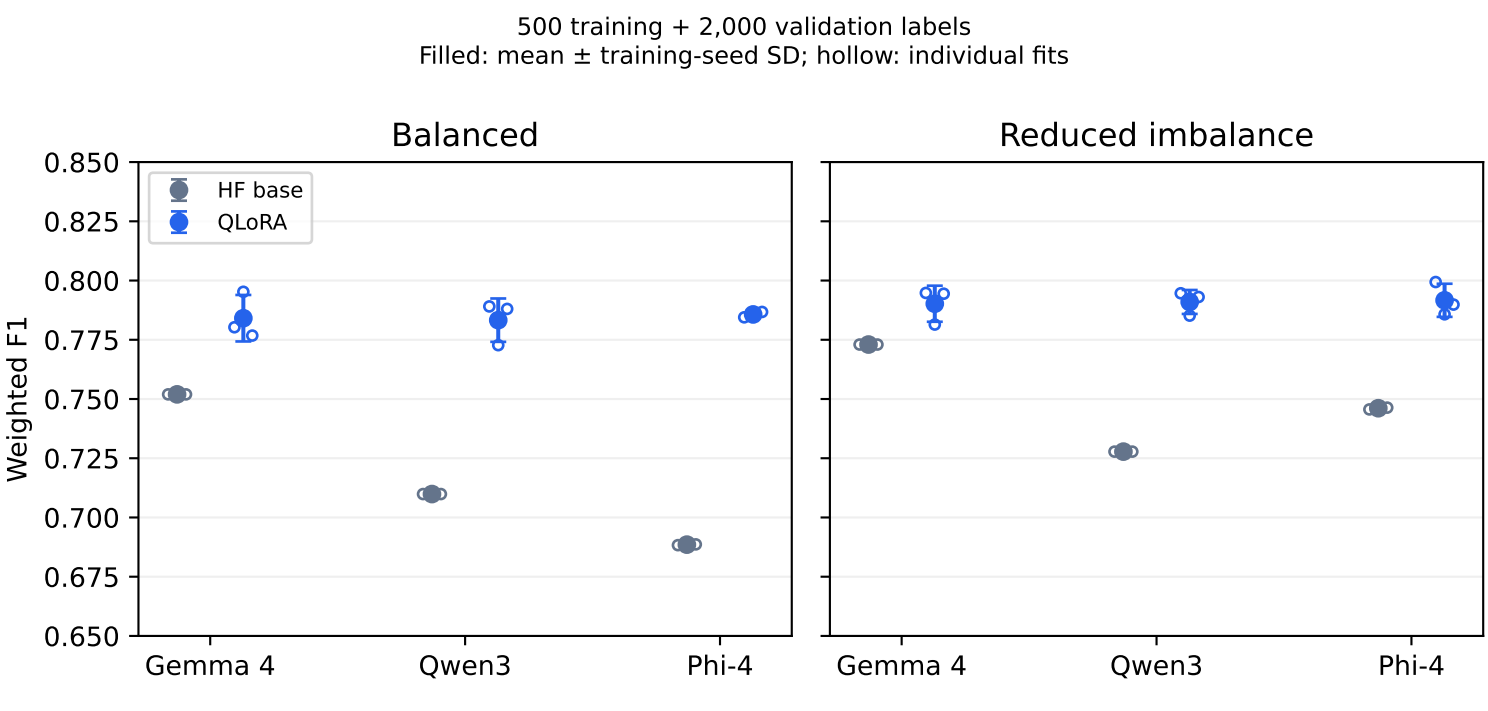
500 training + 2,000 validation labels
Filled: mean ± training-seed SD; hollow: individual fits
Balanced
Reduced imbalance
HF base
QLoRA
Weighted F1
0.850
0.825
0.800
0.775
0.750
0.725
0.700
0.675
0.650
Gemma 4
Qwen3
Phi-4
Gemma 4
Qwen3
Phi-4